6th Annual International IEEE EMBS Conference on Neural Engineering
San Diego, California, 6 - 8 November, 2013

Robust, ultra low-cost MMG system with Brain-Machine-Interface applications

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***Abstract*— Muscle activity is the basis of many brain-machine interface (BMI) applications, but the mainstream EMG-based technology to decode muscle activity has significant constraints when long-term BMI usage "in-the-wild" is required (e.g. controlling neuroprosthetics throughout the day). We use the surface mechanomyogram (MMG), the mechano-acoustic signal generated by lateral oscillations of the muscle fibres during muscle contraction, as source of reliable and robust information of muscle activity. We present our novel MMG sensor and instrumentation, which is designed to match the acoustical properties of muscle signals, while costing at ≈ 10 USD per channel a fraction of current commercial systems. We are able to derive an 'MMG Score' from our sensor-specific signal, which correlates linearly with isometric contraction forces. We test the effectiveness of our MMG system vs EMG using a simple BMI task, where subjects have to interactively control three distinct force states with their muscle activity. Crucially, our MMG Score is robust across subjects, thus calibration on one set of subjects, allows us to predict muscle force production from MMG on other subjects. This limits the need for re-calibration when (re)applying our MMG system to patients on a daily basis, important to minimise carer-dependence and maximise ease of use for BMI users.**

I. INTRODUCTION

A majority of electric prosthesis rely on the user to control them using the electrical signal of their muscles (EMG) [1]. This method is based on non-invasive direct recording of the electrical potentials generated by contracting muscle cells, using electrodes placed on the skin above the muscles. Despite being widely used, the applicability of EMG-based technology has significant constraints in long-term usage. In particular, EMG sensors need precise placement and are highly sensitive to skin impedance changes due to sweat [2]. Another major problem is the cost of low-noise instrumentation and high-gain amplification equipment required to acquire the small electrical signal [3]. A promising alternative is the surface mechanomyogram (MMG), which aims at detecting the mechanical signal generated by lateral oscillations of the muscle fibres during contraction or relaxation. MMG is considered the mechanical counterpart of EMG and it is rich in information on the activity and physical properties of the muscle [4]. Various studies [5], [6] have shown that MMG signal is robust to sensor displacement and, due to its mechanical origin, it is not affected by skin impedance changes. In addition, MMG sensors can be fabricated at lower costs than EMG sensors [7].

Several types of transducers can be used to detect MMG signals, such as accelerometers, microphones, piezoelectric and laser displacement sensors [8]. However, comparisons between them have shown that the shape of the recorded MMG signal changes with each particular sensor used because of the different physical variable measured (i.e. accelerometers measure acceleration whereas lasers measure displacement) [9]. Despite these differences, the major issue in the MMG systems is the interference due to body motion, which can potentially be eliminated by high-pass filtering the signal using a cut-off frequency between 0.1 and 5 Hz [10]. This technique however assumes that the important frequencies components lie in the band-pass of the filter, which is not always true, as the elimination of frequencies below 1 Hz showed significant differences in the resulting signals [11]. Alternative approaches to reject the motion artefact and external noise sources have also been proposed, such as coupled accelerometer-microphone sensor pairs [12] and differential probes [13]. The former shows several technical limitations, as the motion removal scheme relies on the estimation of material properties and the manufacturing process is difficult to be replicated accurately on large scales. The differential probe method, though showing much better rejection of movement artefact, requires more expensive equipment.

MMG has several applications, both in BMI as a control signal for externally-powered prosthetic devices [2], [3], [7], [14] and in neuroscience, medical and biomechanics research, e.g. as a diagnosis tool [15]. Although these studies have examined and tested different approaches to classify movements or limb positions from the MMG signal, no application of regression analysis for MMG-based BMI control has been reported in the literature. Quantitatively determining the applied force through an MMG signal is therefore of fundamental importance for prosthetic device control, something that has only been achieved in EMG-based applications [1] to date.

We present an ultra low-cost design of an MMG sensor and its instrumentation for low-cost BMI applications. More specifically, we designed a circuit for the conditioning and acquisition of the MMG signal and we devised an algorithm to estimate an 'MMG Score' which we show to correlate well with EMG activation across subjects (see Fig. 3). The MMG Score-force relationship is near linear, even when compared across subjects (see Fig. 4), which implies that our MMG Score can be used to roughly predict force production across

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* This research was supported by the EPSRC and the Human Frontiers in Science Program Grant (HFSP RPG00022/2012).

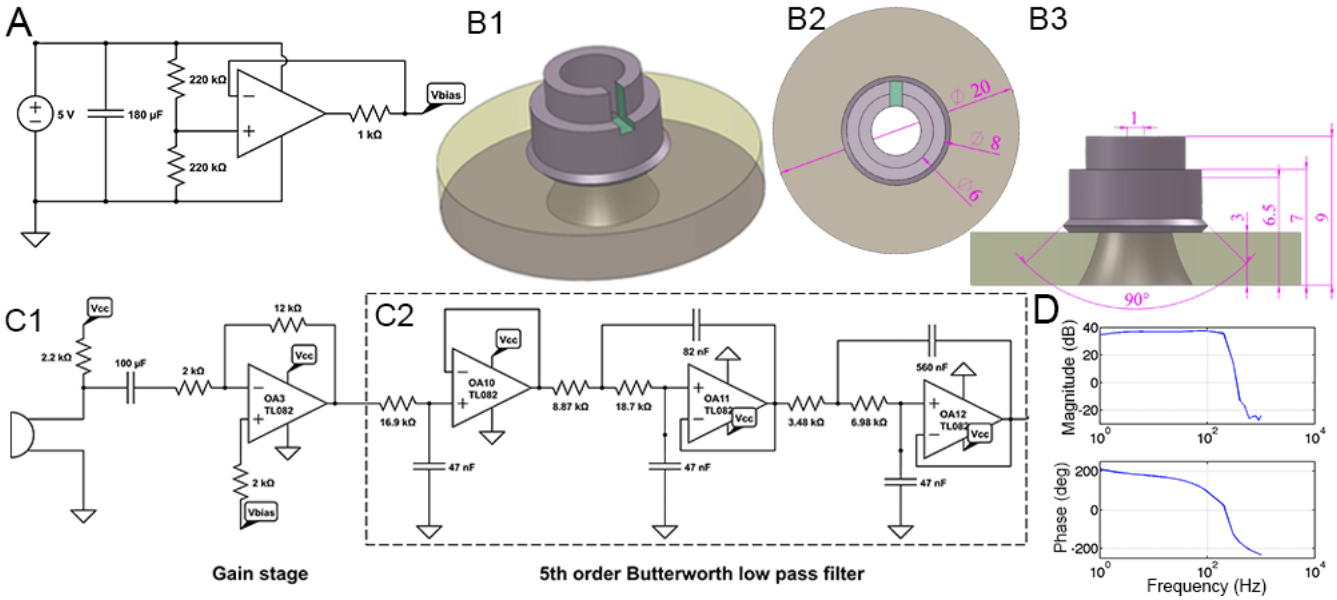

Fig. 1. **(A)** Microphone biasing circuit to provide reference zero-point voltage for microphone signal. **(B1)** Microphone sound chamber with **(B2)** top view and **(B3)** lateral view. All length dimensions shown are in mm. Instrumentation circuit with microphone, **(C1)** gain stage and **(C2)** 5th order low-pass Butterworth filter. The gain stage is connected to the biasing circuit via the V_{bias} connection and provides zero-signal at 2.5 V. **(D)** The bode plot of the instrumentation circuit showing a clear cut-off at 200Hz.

different subjects without needing a per subject calibration. Finally, we apply a pattern-recognition algorithm on our MMG signal in order prove its effectiveness as a control signal in BMI applications.

II. METHODS

Our MMG sensor and instrumentation circuit (see Fig. 1) are based on novel designs aiming at improving signal quality and stability while reducing the manufacturing cost.

MMG sensor design: We use electret condenser microphones for our MMG signal detector, as they have been proven to be robust to body motion artefacts [9]. To get even better mechanical response via the air column, we confined the air chamber between the muscle and the microphone diaphragm (Fig. 1.B2) as to concentrate the pressure waves and transfer the forces generated to the diaphragm more efficiently. The microphone chamber (Fig. 1.B1) has been designed to guarantee a good response to low frequency signals, as the MMG signal frequency band has been shown to lie in the $5-100$ Hz interval[16]. The bottom conical section (Fig. 1.B3) of the chamber ensures that sensor collects mechanical signal from the wider surface. The straight cylindrical section, above the conical section, stabilises the concentrated pressure wave and enhances the sensitivity of the microphone by reducing the effective area. Finally, the rubber section around the chamber area (yellow annular in Fig. 1.B1), seals the chamber to avoid ambient noise whereas the overall tiny sensor size provides higher precision when used on the small muscles.

MMG instrumentation: The MMG instrumentation circuit consists of a microphone with its pre-amplifier (Fig. 1.C1), a low- pass Butterworth filter (Fig. 1.C2) and a data acquisition unit. The pre-amplifier part amplifies the signal collected by the microphone and provides an offset to bias the signal by the half supplied voltage, since the A/D converter has a single voltage reference with precision of 5 mV. Our analog filter provides a frequency cut-off at 200Hz, which preserves the MMG frequency components. Additionally, the 5th order Butterworth filter guarantees a good compromise between high attenuation rate and flat-band pass. We designed the Butterworth filter in Filterlab (Microchip Technology Inc., Chandler, AZ) and the whole circuit was simulated in LTspice (Linear Technology, Milpitas, CA) prior to production.

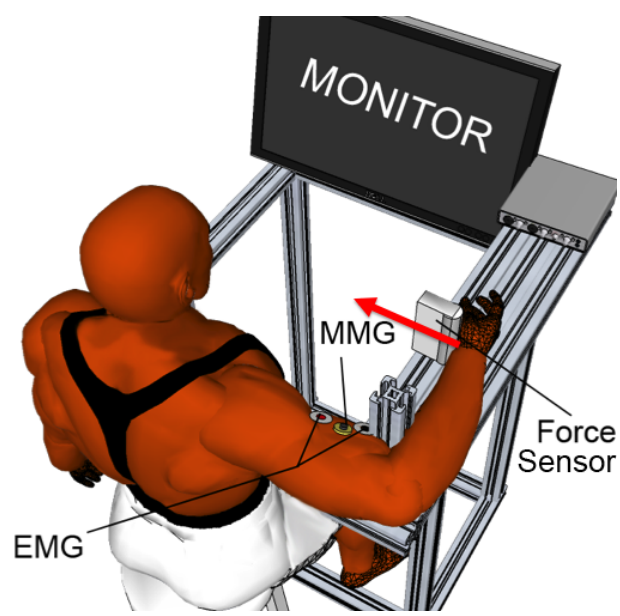

Fig. 2. Experimental setup arrangement showing a healthy subject sitting on a chair, having the elbow at right angle position while resting the arm on an aluminium platform. The force applied is measured using a force transducer and a visual feedback is provided on the monitor. The MMG sensor and EMG electrodes are attached on the belly of the right hand *biceps brachii*.

We tested the frequency response of our instrumentation circuit using ML4856 PowerLab 26T stimulator (AdInstruments, Dunedin, New Zealand) by suppling sinusoidal signals of 500 mV amplitude and $1-1000$ Hz frequency range as input to the circuit. Fig. 1.D shows the frequency response characteristics of our system in a bode diagram, exhibiting a fairly flat band pass with an attenuation of 1.77 dB at 200 Hz. This value deviates from the theoretical 3dB attenuation value as the components used in the circuit design have tolerance. The magnitude of the system response decreased steadily reaching a plateau at around -25 dB after 600 Hz. Likewise, the phase response exhibited a phase shift difference of 200° between 1 Hz and 200 Hz as expected from the LTspice simulation.

III. RESULTS

To validate the accuracy and efficiency of our MMG sensor, 6 healthy male subjects (aged 22-25) were tasked to produce different levels of isometric force in the setup shown in Fig. 2. Each subject was seated on a chair with their right arm placed on a horizontal platform at resting position. At rest the elbow was flexed at 90° and the distal extremity of the forearm was placed in contact with a MLT004/ST Grip force transducer (AdInstruments, Dunedin, NZ) to measure the isometric force of elbow flexion. Our prototype MMG sensor was attached on the bump of the *biceps brachii* muscle using a double sided adhesive tape, along with differential EMG electrodes proximal and distal of our sensor. Subjects performed two simple tasks, discussed in Sections III.A and III.B, for which we recorded force, MMG and EMG signals using a PowerLab 26T (AdInstruments, Dunedin, NZ) at 1 kHz sampling frequency. Data were imported into Matlab (Mathworks Inc., Natick, MA) for further analysis.

A. Comparison between MMG, EMG and muscle force

Each subject was instructed to apply $5, 10, 20, 40, 60, 80$ and 100 N force for 20 s with in-between resting of 20 s.

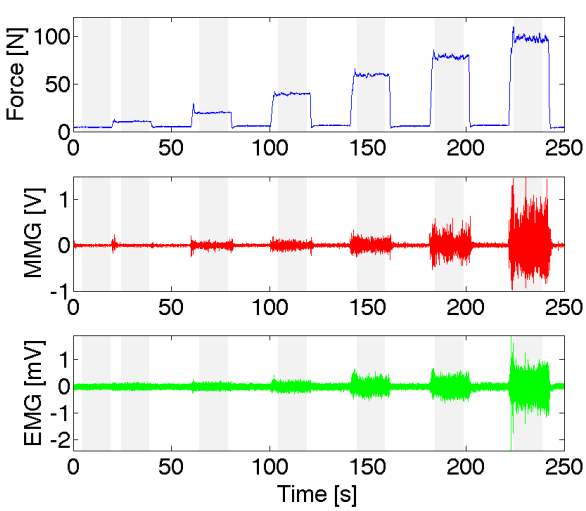

Fig. 3. Isometric force levels with transduced and amplified MMG and EMG timeseries collected from a subject showing the amplitude of both MMG and EMG signals increasing with the applied force. The shaded areas highlight the portion of signal used to compute the MMG Score in Fig. 4.

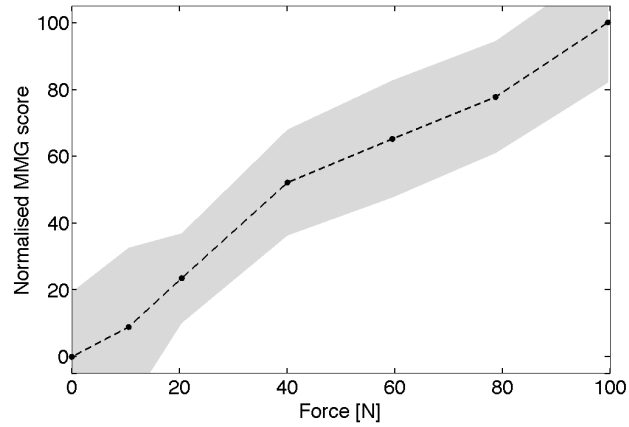

Fig. 4. The normalised average±standard deviation (black dots and dotted line±grey shaded area) of the 'MMG Score' across all subjects (N=6) plotted against the average force levels subjects applied.

The subjects could see their force applied in real-time on a display unit in front of them, which also provided guidance throughout the experiment about the force levels and the resting phases.

The raw signals collected from the MMG and EMG were initially low-pass filtered at both 100 Hz and 300 Hz respectively to remove noise artefacts. Subsequently, the signal time-series were rectified and smoothed to obtain the signal envelope and also the offset was removed. Force, MMG and EMG time-series for a typical subject are shown in Fig. 3, where the amplitude of both MMG and EMG signal clearly increases with the force applied and consequently with level of muscle contraction. Our MMG sensor achieved an 0.84 average correlation coefficient between MMG and EMG signal across all subjects.

The amplitude of the MMG (V) and EMG (mV) signals varied across subjects as expected, since the signals are influenced by individual properties such as muscle mass, muscle length and the thickness of surrounding tissue. We define MMG Score, as $\text{MMG Score} = \text{median}(\log(\text{MMG}_i))$, where i is a 15 second signal collected from our MMG sensor at a specific force level, as shown by the grey shadows in Fig. 3. The collection time of the signal can be varied to much shorter time spans, and was used here to match the time subjects had to hold the force constant. The normalised MMG Score across all six subjects at different contraction levels is shown in Fig. 4, where the increasing magnitude of the mean values correlates with the increasing force level.

B. A simple real-time control BMI application

Three of the subjects were additionally instructed to perform a 3-state control task consisting of a resting phase and two distinct force levels, 40 N and 80 N. For our experiment, subjects performed 10 repetitions of the sequence: Resting - 40 N – Resting – 80 N (each phase lasted 20s). Each MMG and EMG signal envelope was separated into two halves, one to be used as training data and the other as test data. The training data set was constructed by taking non- overlapping data portions of the 1 s length from each muscle state of

control task and assigning them a label representing the state it corresponds to (i.e. performing supervised learning). The transition phases between force levels were rejected as being corrupted by movement artefact. A k-nearest neighbours classifier was used to estimate the instantaneous force state of the other half of the data. In order to do so, the algorithm waits for one second at the beginning and then at each time step it takes the preceding 1 s of data and classifies it.

TABLE I

PERFORMANCE OF EMG VS OUR 'MMG SCORE' IN BMI CONTROL TASK, REPORTED AS % CORRECT DECODING ($N = 6$ SUBJECTS).

Control Step	EMG	MMG
Rest	68.4%	61.4%
40N	92.6%	64.0%
80N	89.4%	95.9%

We report the results of our simple BMI task decoding muscle signals using a commercial EMG and our MMG approach. We show that even our very basic classifier was able to reliably predict the subject's action intention at 80N force even better than the EMG, as at these relatively high forces our MMG signal shows less fluctuations than the EMG. At rest both systems perform approximately twice as good as chance (chance level 33.3%), while at 40N signal the EMG signal still outperforms the MMG system from a limited pool of six subject's data.

IV. CONCLUSIONS

Clinical applications of MMG: Albeit MMG is not a novel technology, it has thus far received only a limited amount of attention, in part because it was difficult to extract a reliable and robust force signal that is applicable across subjects. We have designed an ultra low cost MMG sensor that can be applied in a wide range of BMI applications, such as [17], [14], and integrated with kinematic body sensor networks [18], [19]. We benchmarked our device against a commercial EMG system and showed a distinct relationship between MMG and EMG signals, allowing to reliably predict the produced force levels. Moreover, we demonstrated that subjects are capable of performing a simple BMI control task using our MMG sensor as input, thus confirming the potential use of our system for neuroprosthetic control and rehabilitation. Using our 'MMG Score', we further show that it is possible to use the system directly on a subject which has not been trained before, allowing the technology to be put to immediate use on a new subject.

Our system demonstrated is a continuation of our Frugal Engineering approach to neurotechnology [17], [20], [18], [19], where we aim to use smart algorithms and neurophysiological expertise to overcome expensive hardware components. Therefore, based on our ETHO1 wireless body sensor network technology [18] to collect force and kinematic signals, we designed a system that costs only a fraction of its commercial equivalent EMG but retains the same qualitative and quantitative standards with respect to its output (≈ 10 USD per channel vs $1,000$ USD). This lower cost has the potential of allowing novel applications to develop. The reduction in size and cost granted by our system makes it possible to combine it with wearable motion capture equipment and controlling neuroprosthetics in daily life.

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